Peilin Tan

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Education

University of California, San Diego (UCSD) La Jolla, CA
MS. in Computer Science and Engineering *Sept. 2024 – Mar. 2026 (Expected)*
Current GPA: 4.0/4.0

Wuhan University of Technology (WUT) Wuhan, Hubei
BS. in Information and Computing Science (Computational Mathematics) *Sept. 2018 – Jun. 2022*
GPA: 4.07/5.00 (90.07/100.00)

Research Summary

Multimodal Learning with Large Language Models (LLMs): Designing LLM-enhanced multimodal reasoning network that integrates numerical time series, textual narratives, temporal context, and structural industry information through adaptive hypergraph fusion to achieve domain-aware cross-modal alignment, yielding improved investment returns and robust risk management. [1]

Spatio-temporal and Graph-based Relational Modeling for Time Series Forecasting: Developing spatio-temporal and hierarchical hypergraph-based model that captures dynamic cross-asset dependencies, temporal-causal relations, and higher-order market structures to enable more accurate stock prediction and profitable portfolio optimization. [2]

Publications

[1] **Peilin Tan**, Liang Xie, Churan Zhi, Dian Tu, Chuanqi Shi, "H3M-SSMoEs: Hypergraph-based Multimodal Learning with LLM Reasoning and Style-Structured Mixture of Experts," *under review at WWW 2026, arXiv 2025*.

[2] **Peilin Tan**, Chuanqi Shi, Dian Tu, Liang Xie, "MaGNet: A Mamba Dual-Hypergraph Network for Stock Prediction via Temporal-Causal and Global Relational Learning," *under review at WWW 2026, arXiv 2025*.

[3] Chuanqi Shi, Muhammad Asif Ali, **Peilin Tan**, Huan Wang, "Missing Directions in Directed Relation Learning? Equivalent Projection Alignment Makes It Possible," *under review at AAAI 2026*.

[4] Chuanqi Shi, Asif Ali Muhammad, **Peilin Tan**, Liang Xie, Yulong Wang, Ying Sha, Huan Wang, "Contend-LP: Conflict-Resilient Directed Graph Embeddings for Link Prediction," *under review at WWW 2026*.

[5] Chuanqi Shi, Asif Ali Muhammad, **Peilin Tan**, Liang Xie, Yulong Wang, Ying Sha, Huan Wang, "ARFM: Adaptive Reinforced Filtering Method for Identifying Spurious Links in Noisy Social Networks," *under review at WWW 2026*.

[6] Yechen Yu, Liang Xie, Jiankai Zheng, **Peilin Tan**, "LOSTFormer: Linear Orthogonal Spatio-Temporal Transformer with Learnable Rotation," *under review at DASFAA 2026*.

[7] **Peilin Tan**, "HCA-IDGWO: a Joint Charging Optimization Algorithm for Wireless Rechargeable Sensor Networks," *in ICCT, 2024*.

[8] **Peilin Tan**, "Optimal Design of Maximum SNR for 64-QAM Channel Based on Imperialist Competitive Algorithm," *in CISCE, 2021*.

Research Experience

H3M-SSMoEs: Hypergraph-based Multimodal Learning with LLM Reasoning and Style-Structured Mixture of Experts [1] [arXiv] Jun. 2025 – Oct. 2025
Wuhan University of Technology & UC San Diego | Supervisor: Assoc. Prof. Liang Xie

- Proposed H3M-SSMoEs, a multimodal architecture unifying multi-context hypergraph modeling, LLM-enhanced reasoning, and Style-Structured Mixture of Experts (SSMoEs) for stock prediction across quantitative and textual modalities.
- Designed Multi-Context Multimodal Hypergraph: (1) Local Context Hypergraph capturing fine-grained spatio-temporal dynamics at instance level, and (2) Global Context Hypergraph modeling persistent inter-stock relationships, employing Jensen-Shannon Divergence (JSD) weighting for adaptive hyperedge importance learning.

- Integrated a frozen LLM Llama-3.2 with lightweight adapters to achieve semantic alignment and cross-modal fusion between quantitative market signals and financial news narratives.
- Developed Style-Structured MoEs combining shared market experts and industry-specialized experts with learnable style vectors, enabling regime- and sector-aware specialization.
- Achieved leading risk-adjusted returns across DJIA, NASDAQ 100, and S&P 100 indices with Sharpe Ratios of 1.585 / 2.100 / 1.351 and Calmar Ratios of 3.377 / 4.380 / 2.075, respectively, delivering Annual Returns of 50.00% / 70.80% / 29.62% while maintaining lowest Maximum Drawdowns (14.81% / 16.17% / 14.27%).

MaGNet: A Mamba Dual-Hypergraph Network for Stock Prediction via Temporal-Causal and Global Relational Learning [2] [arXiv] Jan. 2025 – May 2025

Wuhan University of Technology & UC San Diego | Supervisor: Assoc. Prof. Liang Xie

- Proposed MaGNet, a dual-hypergraph architecture integrating temporal-causal and global relational learning for stock movement forecasting.
- Developed the MAGE block, which augments bidirectional Mamba state-space models with adaptive gating, Mixture of Experts (MoEs) for market regime adaptation, and multi-head attention for global context fusion, enabling improved temporal modeling of complex hierarchical financial patterns.
- Designed Feature-wise and Stock-wise 2D Spatiotemporal Attention modules to model feature-to-feature and cross-stock dependencies while preserving temporal structure, bridging temporal modeling with relational reasoning.
- Constructed a dual-hypergraph framework combining Temporal-Causal Hypergraph for fine-grained causal dependencies and Global Probabilistic Hypergraph for market-wide patterns, enabling multi-scale relational learning that disentangles localized temporal influences from global structures.
- Achieved consistently strong performance across 6 major global indices (DJIA, HSI, NASDAQ 100, S&P 100, CSI300, Nikkei 225) with accuracy reaching 54.9%, Sharpe Ratios of 1.05–1.70, Annual Returns up to 22.6% and Calmar Ratios up to 4.36. Maximum Drawdowns remained below 9.3%, demonstrating superior risk-adjusted returns and robust risk management.

Research Assistant Jan. 2023 – Sept. 2024

Wuhan University of Technology | Supervisors: Prof. Shengshuang Chen & Assoc. Prof. Liang Xie *Wuhan, Hubei*

- **Research and Development of Fuel Cell Lifetime Prediction Technology Based on Deep Time Series Models**
  - ∗ Adapted iTransformer for PEM Fuel Cells lifetime prediction, leveraging its inverted dimensional structure to embed multivariate time series and capture degradation-related correlations across operating parameters for accurate 96-step forecasting under limited data scenarios.
  - ∗ Achieved improved performance with only 20% training data (RMSE: 0.00163, R^2: 0.89486, RUL error: 1.0460%) across datasets FC1, FC2, and FC3. Validated computational efficiency with 95% faster training time (2.41s vs. 47.74-74.16s for baselines) and reduced GPU consumption.
- **Federated Learning Project, Peng Cheng Laboratory, PCL-CMCC Foundation for Science and Innovation (Grant No. 2024ZY2B0050)**
  - ∗ Conducted literature reviews on federated learning and explored the integration of spatial and temporal modeling into federated GNN architectures.
  - ∗ Assisted in data processing, experimental design, and technical report writing to support project deliverables.
- Studied advanced sequence and graph neural network models to improve predictive performance via spatio-temporal modeling in time series forecasting and stock prediction. Contributed to research on Swarm Intelligence (SI) optimization algorithms and machine learning.

HCA-IDGWO: A Joint Charging Optimization Algorithm for Wireless Rechargeable Sensor Networks [7] Apr. 2021 – Apr. 2022

Wuhan University of Technology | Supervisor: Prof. Shengshuang Chen

Independent Innovation Research Project, supported by the Fundamental Research Funds for the Central Universities (Grant No. 2021-LX-B1-12)

- Proposed HCA-IDGWO, a hybrid Swarm Intelligence (SI) algorithm integrating Hybrid Clustering Analysis (HCA) and Improved Discrete Grey Wolf Optimization (IDGWO) to jointly optimize UAV deployment, network partitioning, and charging trajectory planning in multi-UAV Wireless Rechargeable Sensor Networks (WRSNs).
- Designed the HCA module by combining hierarchical clustering with improved K-means to partition large-scale WRSNs and balance UAV workloads, enabling adaptive deployment and allocation of UAVs.
- Developed the IDGWO framework by discretizing the Grey Wolf Optimizer with two-part encoding representation and incorporating Crossover Operator and Large Neighborhood Search (LNS) with destroy-repair operators to achieve improved exploration-exploitation balance and accelerated convergence.

- Conducted experiments on 18 TSPLIB benchmarks with rigorous statistical tests (Friedman Test p-value = 3.0054e-7, Nemenyi Post Hoc, One-Way ANOVA, Bonferroni Test), achieving an average error deviation of 1.69% and demonstrating superior accuracy, robustness, and convergence over baselines while maintaining the shortest charging distances, thereby minimizing UAV flight energy consumption and ensuring network-wide energy balance.

Deep Discrete Hashing Based on Intelligent Optimization and Its Application in Image Retrieval
Wuhan University of Technology | Supervisors: Prof. Shengshuang Chen & Assoc. Prof. Liang Xie
Excellent Graduation Thesis *Dec. 2021 – Jun. 2022*

- Designed a Deep Discrete Hashing (DDH) network integrating deep feature extraction and Swarm Intelligence (SI) optimization for efficient and accurate image retrieval.
- Adopted ResNet-18 as the backbone for feature extraction, employed a Cauchy-based pairwise cross-entropy loss, and applied transfer learning to generate compact, discriminative hash codes for images.
- Developed a hybrid Swarm Intelligence optimization algorithm, combining Cuckoo Search and Grey Wolf Optimizer with Opposition-Based Learning (OBL-CSGWO) to optimize discrete hash code generation and enhance convergence stability.
- Conducted experiments and sensitivity analyses on the CIFAR-10 dataset, achieving mAP@50 of 91.28%, mAP@H $\leq$ 2 of 77%, and a recall rate of 88.0%, demonstrating high retrieval accuracy while validating the effectiveness of integrating Swarm Intelligence strategies into deep hashing network for enhanced image retrieval performance.

Industrial Experience

Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences (SIAT, CAS)
Technician | Brain Connectivity Mapping Group | The Brain Cognition and Brain Disease Institute | Supervisors: Assoc. PI Yanyang Xiao & Assoc. PI Fang Xu Jul. 2022 – Jan. 2023

- Designed a 3D Spatial Alignment Network (3D-SAN) for precise automatic registration of whole-brain medical images, improving alignment accuracy through an enhanced U-Net architecture that integrates Spatial Transformer Networks (STN) and a novel Alignment Transform Network (ATN), achieving sub-pixel registration precision on 3D MRI data from the LPBA40 dataset.
- Assisted in developing deep learning models for the project "Projection Map of the Lateral Geniculate Nucleus", reconstructing large-scale 3D brain images from high-throughput monkey brain data.
- Leveraged GPU clusters to process TB–PB level optical microscopy imaging data acquired with the VISoR (Volumetric Imaging with Synchronized on-the-fly-scan and Readout) 3D imaging system.

Hikvision Digital Technology Co., Ltd.
Application Software Development Engineer Intern | Shared Software Research and Development Center Jul. 2021 – Sept. 2021

- Gained hands-on experience with the Linux system, including Bash scripting, regular expressions, and Vim for file management, text processing, and automation tasks. Developed and debugged Shell scripts for automation and process optimization.
- Assisted in maintaining software systems by applying patches, automating operational workflows with Shell scripts, and contributing to software development and optimization projects.

Honors & Awards

2023 Mathematical Contest in Modeling of America (MCM) – *Meritorious Winner (Top 7%)*
2020 "Shenzhen Cup" Mathematical Modeling Challenge – *Third Prize (Top 1%)*
2021 Asia and Pacific Mathematical Contest in Modeling (APMCM) – *First Prize (Top 5%)*
2021 "Central China Cup" Mathematical Contest in Modeling – *First Prize (Top 5%)*
2021 "CSEE Cup" National Electrical Mathematical Modeling Competition – *Third Prize*
2020 Mathematical Contest in Modeling, Wuhan University of Technology – *Second Prize*
Second-Class Scholarship, Wuhan University of Technology
Academic Research Scholarship, Wuhan University of Technology
Excellent Graduation Thesis, Wuhan University of Technology
Reviewer, *AAAI Conference on Artificial Intelligence 2026*

Technical Skills

Languages: Python, Matlab, C/C++, Linux Shell, Latex
English Proficiency: IELTS: 7.0, GRE: 328 + 3.5 (Quantitative: 169, Verbal: 159, Writing: 3.5)
Machine Learning Tools: PyTorch, Hugging Face Transformers